



React Full Stack (MERN) Developer Program

Build production-grade web apps end-to-end with MongoDB, Express.js, React.js and Node.js — from UI components to deployed REST APIs.

LEARNING PATH · PROGRAM SYLLABUS · REAL-TIME PROJECTS & CASE STUDIES

8 Months

Beginner to Job-Ready

4

MERN Technologies

8+

Real-World Projects

850+

Hiring Partners

01 Learning Path

1

Web Fundamentals, Git & JavaScript Core

Month 1

HTML5, CSS3, Flexbox & Grid, JavaScript ES6+ (DOM, functions, arrays, objects), VS Code & DevTools — capped with a responsive portfolio site.

2

React Mastery (Frontend Pro Dev)

Month 2

React components, props, state & lifecycle; Hooks (useState, useEffect, useContext, useReducer); routing & conditional rendering — building a Job Portal UI with form validation.

3

Backend Engineering — Node.js & Express

Month 3

Node.js core concepts, Express.js routing & middleware, REST API development, Postman/Thunder Client, and JWT/Bcrypt authentication.

4

Database & Full-Stack Integration

Month 4

MongoDB CRUD & Mongoose, SQL basics for interviews, environment variables, CORS & error handling — building a full-stack blog platform with auth.

5

DevOps, Deployment & Testing

Month 5

Docker & Docker Compose, GitHub Actions CI/CD, AWS EC2/S3, Vercel/Netlify deployment, and unit testing fundamentals with Jest.

6

Capstone Projects & Placement Preparation

Months 6–8

Build an e-commerce platform with admin dashboard and an LMS clone, then complete an AI-powered capstone project alongside mock interviews and placement-focused DSA practice.



02 Program Syllabus

TERM 1 · FRONT-END FOUNDATIONS

FRONTEND

HTML5, CSS3 & Modern JavaScript

- Semantic HTML5, Flexbox & CSS Grid
- JavaScript ES6+ — DOM, functions, arrays, objects
- Async/await, Promises & the Fetch API
- Responsive, mobile-first design

React.js Fundamentals

- JSX syntax and functional vs. class components
- Props, state and component lifecycle methods
- React Hooks — useState, useEffect, useContext
- Conditional rendering, lists & keys

TOOLS & TECHNOLOGIES

HTML5

CSS3

JavaScript ES6+

React.js

TERM 2 · ADVANCED REACT & STATE MANAGEMENT

FRONTEND
PRO

Routing & Styling

- React Router — setup, routes & parameters
- Nested routes and dynamic routing
- CSS Modules, CSS-in-JS & styling libraries
- Tailwind CSS for UI/UX development

Redux & API Integration

- Redux setup — actions, reducers & store
- Connecting components to the Redux store
- Asynchronous programming & API integration
- Error handling, debugging & performance optimisation

TOOLS & TECHNOLOGIES

React Router

Redux

Tailwind CSS

TERM 3 · BACKEND ENGINEERING — NODE.JS & EXPRESS

BACKEND

Node.js & Express.js

- Node.js core concepts — modules, FS, async patterns
- Express.js routing & middleware
- Building and testing REST APIs
- Postman and Thunder Client workflows

Authentication & Security

- JWT-based authentication
- Password hashing with Bcrypt
- Environment variables & CORS handling
- Centralised error handling middleware

TOOLS & TECHNOLOGIES

Node.js

Express.js

JWT

Postman



TERM 4 · DATABASE, DEVOPS & DEPLOYMENT

DATABASE
+ DEVOPS

MongoDB & Mongoose

- MongoDB CRUD operations & relations
- Mongoose schema design and data modelling

- SQL fundamentals for technical interviews
- Database indexing & query performance

DevOps & Cloud Deployment

- Docker & Docker Compose containerisation
- GitHub Actions CI/CD pipelines

- AWS EC2, S3, Vercel & Netlify deployment
- Unit testing fundamentals with Jest

TOOLS & TECHNOLOGIES

MongoDB

Mongoose

Docker

AWS

Vercel



03 Real-Time Projects & Case Studies

Job Portal UI with Form Validation

Project

A React-driven job-listing interface with multi-field form validation and dynamic state handling.

Hands-on Build

User Auth API + Dashboard

Project

A secure Express REST API with JWT authentication powering a protected user dashboard.

Hands-on Build

Blog Platform with Comments

Project

A full-stack blogging app with user authentication, posts and a nested comment system on MongoDB.

Hands-on Build

Hospital Management System

Project

A single-page app using a JSON-server backend where doctors are added by specialisation and patients book appointments.

Hands-on Build

Online Banking Application

Project

Registration, login, deposit, withdrawal and fund-transfer features for everyday banking workflows.

Hands-on Build

Online Shopping Platform + Admin

Capstone

Full e-commerce app — auth, catalog, cart, order tracking and Razorpay payments, plus an admin dashboard for inventory & sales analytics.

Capstone Project

Interactive LMS Clone

Capstone

A learning platform with course enrolment, video lessons, assignment submission, grading and real-time mentor chat.

Capstone Project

Team Collaboration & Task Tool

Project

A Jira/Trello-style Kanban tool with drag-and-drop tasks, file sharing and real-time notifications.

Hands-on Build

AI-Based Career Mentor Chatbot

Project

A MERN + LangChain/OpenAI chatbot that analyses resumes, suggests improvements and recommends job openings.

Hands-on Build